

成功安装环境和相关软件

```
run@zhi:~/Compile/log/0-dl$ gcc --version
gcc (Ubuntu 9.4.0-1ubuntu1~20.04.2) 9.4.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
run@zhi:~/下载$ bison --version
bison (GNU Bison) 3.5.1
由 Robert Corbett 和 Richard Stallman 编写。
```

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可售性或适用于任何特定目的的保证。

```
run@zhi:~/下载$ flex --version
flex 2.6.4
```

The screenshot displays the QtSPIM simulator window. The top menu bar includes File, Simulator, Registers, Text Segment, Data Segment, Window, and Help. Below the menu is a toolbar with icons for file operations, simulation control, and viewing options. The main window is divided into two panes. The left pane, titled 'Int Regs [16]', shows the state of the processor registers: PC=0, EPC=0, Cause=0, BadVAddr=0, Status=3000ff10, HI=0, LO=0, R0-R15 with their corresponding MIPS register names and values. The right pane, titled 'Text', displays assembly code for two segments: 'User Text Segment [00400000]..[00440000]' and 'Kernel Text Segment [80000000]..[80010000]'. The assembly code includes instructions like lw, addiu, sll, addu, jal, nop, ori, syscall, addu, lui, sw, mfc0, srl, and andi, each with its address and a comment. At the bottom of the window, a status bar indicates 'Memory and registers cleared' and provides version information: 'SPIM Version 9.1.24 of August 1, 2023 (final) Copyright 1990-2023 by James Larus. All Rights Reserved. SPIM is distributed under a BSD license. See the file README for a full copyright notice. QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.'

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